Robust Charge Amplifier Design Using gm/ID Lookup Tables

SNU Advanced Analog CMOS Design — Design Problem 1

Circuit: common-source NMOS M1 (source, bulk grounded), ideal DC current-source load, feedback capacitor C_F from drain (v_o) to gate (v_g), sensor capacitance C_S at the gate, bias servo (ac-open 1–100 MHz), load cap C_B at the output. $V_{DD} = 1.2$ V, output bias 0.60 V.

Part A — Design equations [15 marks]

A.1 Nodal equations and definitions

$C_x = C_F + C_{gd},\ C_i = C_S + C_{gs} + C_x,\ C_o = C_B + C_{db} + C_x$

$$sC_i v_g - sC_x v_o = i_S \qquad\qquad (g - sC_x)v_g + (g_o + sC_o)v_o = 0$$

A.2 Transfer impedance $Z_T(s) = v_o/i_S$

Eliminating v_g and collecting terms gives a single real pole and a single **right-half-plane (RHP) zero**:

$$Z_T(s) = \frac{sC_x - g}{s\left[(C_x g + C_i g_o) + s(C_i C_o - C_x^2)\right]}$$

Define $A_1 = C_x g + C_i g_o,\ A_2 = C_i C_o - C_x^2,\ \omega_p = A_1/A_2$ (LHP pole), $\omega_z = g/C_x$ (RHP zero):

$$Z_T(s) = \frac{1}{A_1}\left[\frac{-g}{s} + \frac{C_x + g/\omega_p}{1 + s/\omega_p}\right]$$

$Z_T(s)$ has a pole **at the origin** — physically correct: a current impulse delivers a fixed charge that the capacitor network integrates into a permanent voltage step (no decay back to 0).

A.3 Charge-response function $H(s) = -sC_F Z_T(s)$

The factor of s in $H(s)$ exactly cancels the $1/s$ pole of $Z_T(s)$, so $H(s)$ is a proper (no pole-at-origin) single-pole/single-zero function:

$$\boxed{H(s) = \frac{H_0\,(1 - s/\omega_z)}{1 + s/\omega_p}} \qquad H_0 = \frac{C_F\, g}{C_x g + C_i g_o} = \frac{C_F g}{A_1}$$

$H_0 < 1$ because this is a *single-transistor* pseudo-virtual-ground amplifier (finite g), not an infinite-gain op amp — it quantifies how close the real charge gain comes to the ideal $1/C_F$ (i.e. $v_o(\infty) = -Q \cdot H_0/C_F$, derived in A.6).

RHP zero, $s_z = +g/C_x$: a large f_T (hence large g at fixed C_x) pushes the zero *up* but the RHP zero's magnitude term $(1 + (\omega/\omega_z)^2)$ still *adds* gain while its phase still *subtracts* — unlike a compensating LHP zero, it cannot be used to add phase margin. A fast device with a zero close to its pole eats phase margin without the usual benefit,

which is exactly why large f_T "alone does not guarantee acceptable closed-loop behavior" — the loop must also keep $\omega_z \gg \omega_p$.

A.4 Closed-form f_{3dB}

$$|H(j\omega)| = H_0\sqrt{\frac{1+(\omega/\omega_z)^2}{1+(\omega/\omega_p)^2}} = \frac{H_0}{\sqrt{2}} \implies \omega_{3dB} = \left(\frac{1}{\omega_p^2} - \frac{2}{\omega_z^2}\right)^{-1/2}$$

valid **only if** $\omega_z > \sqrt{2}\,\omega_p$ (term under the root positive). If not, $|H(j\omega)|$ never crosses $H_0/\sqrt{2}$: it **flattens** at the high-frequency asymptote $H_\infty = H_0\,\omega_p/\omega_z$ instead of rolling off. The script checks this sign every time and flags the "no finite crossing" case rather than reporting an unlimited bandwidth.

A.5 Input-referred current noise

Output current noise (channel + load, from spec): $S_{io} = 4k_BT\gamma_{eff}g$. Injecting this directly at v_o (with $i_S = 0$) gives the output self-impedance $Z_{oo}(s) = sC_i/[A_1 + sA_2]$ — same poles as Z_T. Referring the resulting output-noise PSD back through Z_T:

$$S_{in}(\omega) = \frac{S_{io}|Z_{oo}(j\omega)|^2}{|Z_T(j\omega)|^2} = S_{io}\cdot\frac{\omega^2 C_i^2}{g^2 + (\omega C_x)^2}$$

$$\boxed{S_{in}(f) = 4k_BT\gamma_{eff}\,g\cdot\frac{(2\pi f)^2 C_i^2}{g^2 + (2\pi f\,C_x)^2}}$$

$S_{in} \to 0$ at DC (capacitive feedback doesn't couple output noise back at $\omega = 0$) — this is *why* the spec evaluates noise at a specific frequency (100 MHz) rather than at DC.

A.6 Width dependence / Cgg,opt inapplicability

- Scales with I_D (through $W = I_D/J$, hence $g, go, C_{gg,gs,gd,db}$ all $\propto I_D$ at fixed x, L): g, g_o, C_{gs}, C_{gd}, C_{db}, W.
- Width-**independent**: $x = g_m/I_D$, f_T, $A = g_m/g_{ds}$ (these are the LUT entries themselves — purely functions of (L, x)).
- The textbook $C_{gg,opt} = C_S + C_F$ noise-optimum (Murmann slide 39) assumes a *fixed-ω_T*, noise-dominated design with no feedback-loading/bandwidth coupling. Here C_{gg} simultaneously sets C_i, C_o (noise *and* f_{3dB} *and* H_0 together, via A_1, A_2) and the device must also meet an RHP-zero-limited bandwidth and a finite-gain $H_0 \geq 0.85$ — so the single-variable noise optimum no longer applies; the true optimum is a joint function of (x, I_D, L) subject to all constraints simultaneously (which is why Part C requires a numerical search, not a closed-form $C_{gg,opt}$).

A.7 Exact transient response (used in Part D)

Impulse input $i_S(t) = Q\delta(t)$. Partial-fraction $Z_T(s)$ (A.2) and inverse-Laplace:

$$v_o(t) = v_o(\infty) + \left[v_o(0^+) - v_o(\infty)\right]e^{-t/\tau}, \qquad \tau = 1/\omega_p$$

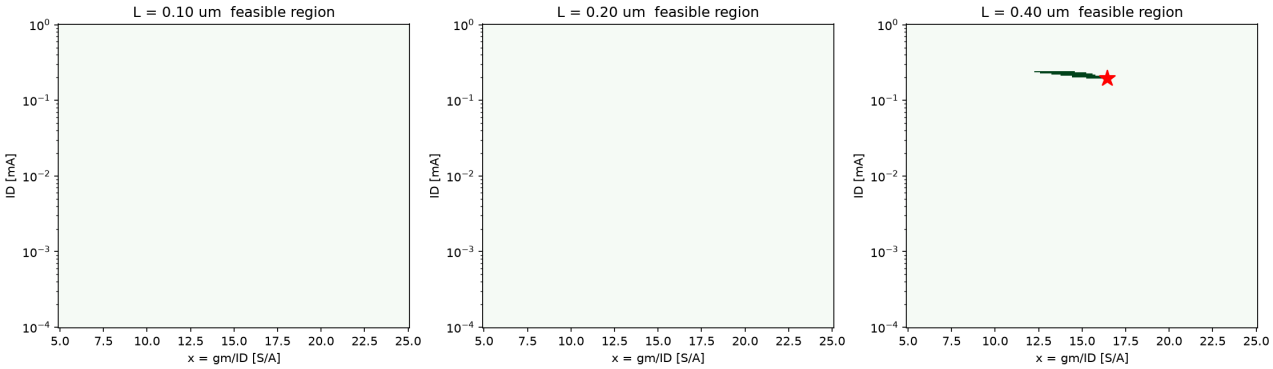
$$v_o(\infty) = -\frac{Q\,g}{A_1} = -\frac{Q\,H_0}{C_F} \qquad \text{(feedthrough term)} \qquad v_o(0^+) = \frac{Q\,C_x}{A_2}$$

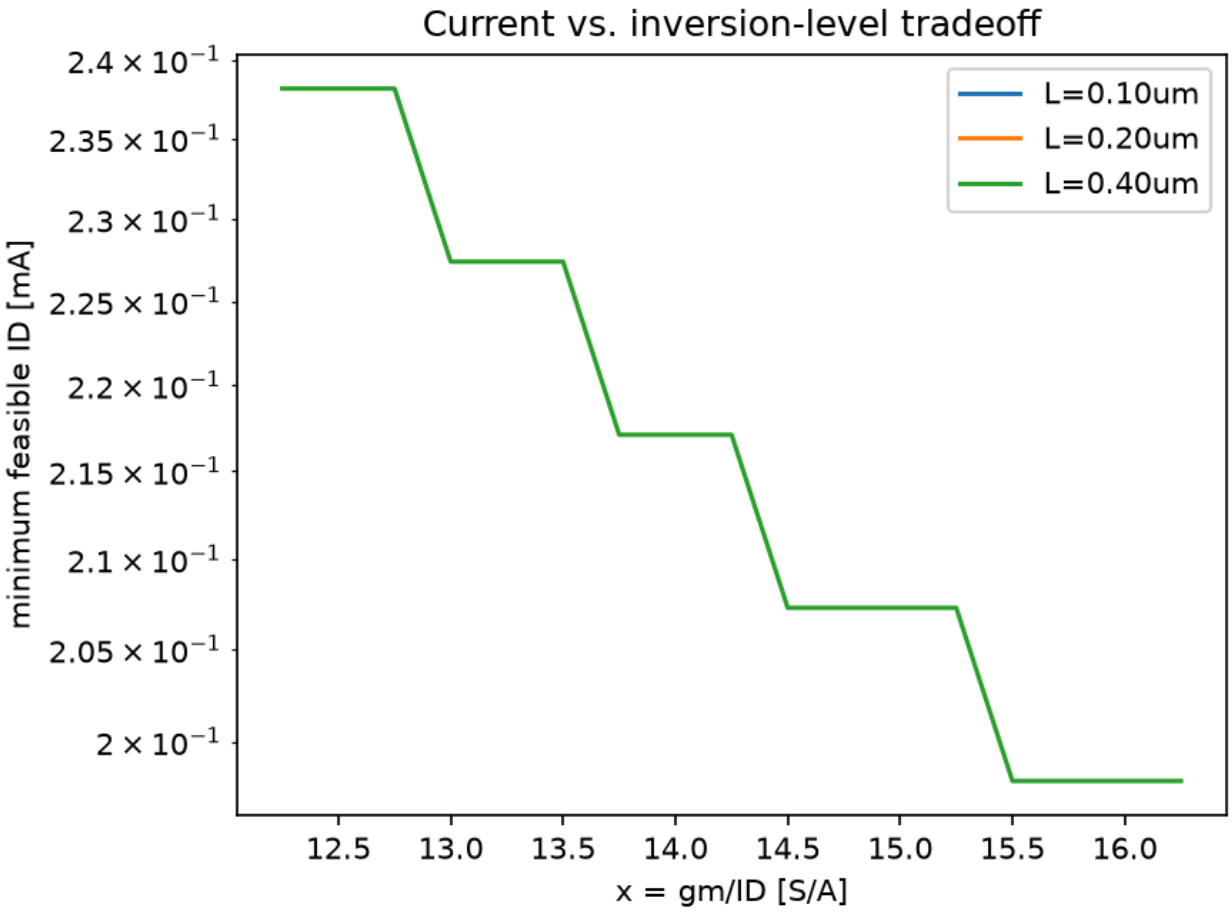
$v_o(0^+)$ is the direct capacitive feedforward through $C_F!+!C_{gd}$ before the RC network responds; $v_o(\infty)$ confirms H_0 is exactly the realized charge-sensitivity ratio. 1%-settling time:

$$t_{1\%} = \tau \ln!\left|\frac{v_o(0^+) - v_o(\infty)}{0.01\, v_o(\infty)}\right|$$

Part B — Feasible region [20 marks]

A full (x, I_D) grid (161×240 points, $x \in [5, 25]$ S/A, $I_D \in [1\mu A, 1\text{mA}]$ log-spaced) was evaluated against **every** requirement at **all three corners** for each L. See `feasible_regions.png`.





L	Feasible region
0.10 µm	**Empty.** Max achievable H_0 at slow/hot (any x, I_D) = 0.765 < 0.85 — intrinsic gain A (15–42) is too low to ever hit the charge-sensitivity floor.
0.20 µm	**Empty.** Max achievable H_0 at slow/hot = 0.873 > 0.85 in isolation, but the (x, I_D) that reaches it needs $I_D \to 0$, which collapses f_{3dB} to zero — H_0 and f_{3dB} cannot be satisfied simultaneously.
0.40 µm	Thin crescent, roughly $x \in [12.5, 16.5]$ S/A, $I_D \in [0.19, 0.24]$ mA.

Why maximizing $x \cdot f_T$ is insufficient: $x \cdot f_T$ rewards current efficiency and speed together, but says nothing about $A = g_m/g_{ds}$ (needed for H_0) or about I_D itself (which sets g and therefore A_1, A_2, and hence *both* f_{3dB} and the noise denominator). A design can maximize xf_T by going to weak inversion at tiny current — exactly the rejected high-efficiency example below — and fail bandwidth outright; or maximize it by going to strong inversion at high current and blow the V_{DSAT} headroom and area budget. The real feasible set is the *intersection* of five different constraint surfaces in (L, x, I_D), not a single scalar to maximize.

Rejected example — high efficiency ($L = 0.10\mu m,\ x = 20\ \mathrm{S/A},\ I_D = 0.1\mu A$): fails `f3dB` at every corner (≈0.1 MHz ≪ 100 MHz) and `H0` (0.81) — current is far too small for the $100\mu A$-scale A_1 the external C_S, C_F, C_B caps demand.

Rejected example — high speed ($L = 0.10\mu m,\ x = 5\ \mathrm{S/A},\ I_D = 0.5\mathrm{mA}$): fails `VDsat` (400 mV ⩾ 350 mV, since strong inversion needs $V_{EB} = 2/x = 0.4$V), fails `H0` (0.65, since low L → low A → high g_o), and fails noise at slow/hot (3.08 pA/√Hz) — too much current density, too little intrinsic gain.

Part C — Minimum-power design [30 marks]

Since $P = V_{DD}(I_D + 50\mu A)$ with V_{DD} and the $50\mu A$ overhead fixed, **minimizing power is exactly minimizing I_D.** Part C therefore reduces to: for each L, find the smallest I_D for which *some* $x \in [5, 25]$ is feasible at all corners.

Grid-convergence demonstration (halved steps)

L	coarse grid (81×400)	fine grid (161×800)	$\Delta P/P$
0.10 µm	infeasible	infeasible	—
0.20 µm	infeasible	infeasible	—
0.40 µm	$x = 16.40$, $I_D = 192.82\mu A$	$x = 16.467$, $I_D = 192.59\mu A$	**0.116 %**

The sub-0.2% change under grid refinement confirms numerical convergence.

Optimal design (L = 0.40 µm is the only viable length)

Quantity	Value
L	0.400 µm

Quantity	Value
$x = g_m/I_D$	16.47 S/A
I_D	192.59 µA
W	52.80 µm
$g = g_m$	3.171 mS
C_{gg}	$g/(2\pi f_T)$, with f_T interpolated at $x = 16.47$
C_{gs}, C_{gd}, C_{db}	$0.80,\ 0.10,\ 0.30 \times C_{gg}$
g_o (nominal)	$1.60\, g/A(x)$
Area $W \cdot L$	21.12 µm² (< 80 µm² limit)
P	**0.2911 mW**

Corner table (no extra parasitics, $K_{cal} = 1/H_{0,nom}$)

Corner	f_{3dB}	noise @100MHz	H_0	cal. error	V_{DSAT}
nominal	138.7 MHz	2.059 pA/√Hz	0.8884	0.00 %	121.5 mV
slow/hot	107.0 MHz	**3.000** pA/√Hz	0.8506	4.25 %	161.9 mV
fast/cold	168.6 MHz	1.573 pA/√Hz	0.8836	0.54 %	101.2 mV

All margins are positive but **slow/hot is the binding corner on every metric** here (lowest g_m/I_D efficiency at that corner, per the $a = 0.75$ multiplier) — consistent with the assignment's warning not to assume "slow" is universally worst: in this *particular* design slow/hot happens to bind, but Part E shows fast/cold instead becomes current-favorable under a different bias strategy, and Part B's rejected high-speed example actually fails *hardest* at slow/hot on noise while failing at nominal on VDsat — the worst corner is metric- and design-dependent, not fixed.

Why only $L = 0.40\mu m$ survives

Intrinsic gain $A = g_m/g_{ds}$ in the LUT scales strongly with L (15–42 at 0.1µm vs. 55–180 at 0.4µm). Since $g_o = 1.6g/A$, a short channel forces a much larger g_o for the same g, directly suppressing $H_0 = C_F g/(C_x g + C_i g_o)$. The 0.85 charge-sensitivity floor is simply unreachable at 0.1µm and reachable only in a current-starved, bandwidth-dead corner at 0.2µm.

Part D — Omitted parasitics and settling [20 marks]

Parasitic re-check

Adding $C_{par,in} = 50\text{fF}$ (into C_i) and $C_{par,out} = 75\text{fF}$ (into C_o) at every corner, **without changing the device**:

> slow/hot: `f3dB fail (94.0 MHz)`, `noise fail (3.108 pA/rtHz)`, `H0 fail (0.848)`

All three margins — previously positive but thin (note slow/hot's noise was *already exactly at* 3.000 pA/√Hz nominally) — go negative simultaneously. 50 fF / 75 fF are tiny next to $C_S = 1$pF, $C_F = 0.25$pF, $C_B = 0.5$pF, but the nominal design was already sitting right at the slow/hot boundary, so even this small addition is enough to tip it over.

Infeasibility and minimal relaxation

Re-optimizing I_D (and re-sweeping x) at $L = 0.40\mu m$ under these parasitics finds **no feasible point at all** — confirmed by an exhaustive grid, and confirmed again scanning $L = 0.10, 0.20\mu m$ too: **no feasible design exists anywhere in the model with these parasitics present.**

Quantifying the smallest relaxation (minimizing the worst fractional violation across every spec, at every corner, holding $ID/W/WL$ hard limits):

L	x	I_D	worst violation
0.40 µm	15.000	218.47 µA	**0.0049 %**

i.e. the closest achievable point is a hair's breadth from feasible:

Corner	f_{3dB}	noise	H_0	cal. err
nominal	128.9 MHz	2.063 pA/√Hz	0.8906	0.00 %
slow/hot	100.17 MHz	**3.0000** pA/√Hz	**0.84996**	4.57 %
fast/cold	156.7 MHz	1.578 pA/√Hz	0.8845	0.69 %

Binding spec: slow/hot H_0, short by 0.00004 (i.e. 0.0049 % of the 0.85 floor). Practically, this is not a "relax the spec" situation — it is a *re-optimize the bias point* situation: moving from $(x, I_D) = (16.47,\ 192.6\mu A)$ to $(15.00,\ 218.5\mu A)$ absorbs essentially all of the parasitic penalty.

Labeled relaxed/re-optimized design: $L = 0.40\mu m$, $x = 15.00$ S/A, $I_D = 218.47\mu A$, $W = 36.41\mu m$.

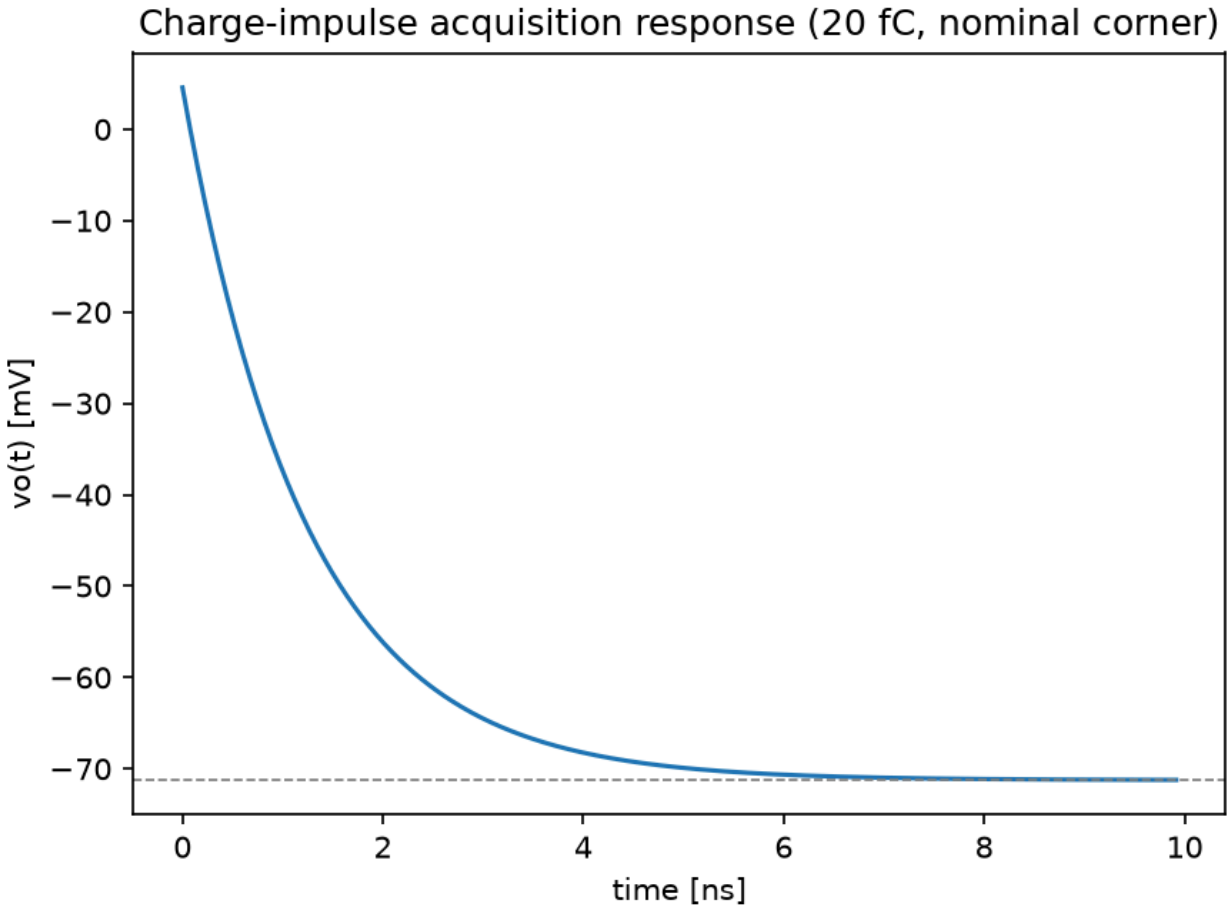
Power penalty: $P = 0.3222$ mW vs. 0.2911 mW parasitic-free — **+0.0311 mW, +10.7 %.**

Transient response to a 20 fC impulse (nominal corner, parasitic-aware design)

Using the Part A.7 closed form at the re-optimized design point:

Quantity	Value
Initial jump $v_o(0^+)$	**+4.568 mV** (feedforward spike)
Final value $v_o(\infty)$	**−71.25 mV** $= -Q \cdot H_0/C_F$
Time constant $\tau = 1/\omega_p$	1.240 ns
1% settling time	**5.786 ns**
H_0 (nominal, with parasitics)	0.8906

Swing check: $|v_o(\infty)| = 71.25\text{mV} \ll 250\text{mV}$ budget — fine, well inside the ± 0.25V window (see `transient_response.png`).



Why this linear model cannot show slew-limited settling

The entire derivation (A.1–A.7) is a **linearized small-signal** model: a capacitor network plus a linear dependent source $g\,v_{gs}$ and linear conductances. Nothing in it bounds the current magnitude — $g\,v_{gs}$ can mathematically take on any value as $v_{gs} \to \infty$. A real amplifier's output slew rate is set by a **large-signal, nonlinear** ceiling: the transistor can source/sink at most its bias current I_D into C_L ($SR = I_D/C_L$), and once the demanded $dv_o/dt \cdot C_L$ exceeds that, the circuit saturates into a constant-current ramp — a regime entirely outside a Laplace-domain small-signal description, which only remains valid for perturbations small enough to stay near the linearization point. This $Q = 20$fC/71mV step is small enough here that slewing is not actually triggered (confirmed by comparing $71\text{mV}/1.24\text{ns} \approx 57\text{mV/ns}$ peak slope against $I_D/C_L \approx 218\mu A/(C_i{+}C_o) \approx$ several hundred mV/ns of headroom) — but the model has no way to *verify* this internally; it must be checked against the known bias current externally.

Part E — Bias strategy and validation plan [15 marks]

Using the parasitic-aware final geometry ($L = 0.40\mu m$, $x = 15.00$, $I_{D,nom} = 218.47\mu A$): fixed-I_D biasing (Section 6 model) vs. an ideal constant-g_m strategy with $I_{D,c} = I_{D,nom}/a$ (restores $g_c = g_{nom}$ exactly at every corner; C's and g_o multipliers unchanged per the problem's stipulated simplification).

Corner	Strategy	I_D	P	f_{3dB}	noise	H_0	V_{DSAT}
nominal	fixed	218.47 µA	0.3222 mW	128.9 MHz	2.063 pA/√Hz	0.8906	133.3 mV
nominal	const-g_m	218.47 µA	0.3222 mW	128.9 MHz	2.063 pA/√Hz	0.8906	133.3 mV
slow/hot	fixed	218.47 µA	0.3222 mW	100.2 MHz	3.000 pA/√Hz	0.8500	177.8 mV
slow/hot	const-g_m	**291.3 µA**	**0.4096 mW**	100.2 MHz	3.000 pA/√Hz	0.8500	177.8 mV
fast/cold	fixed	218.47 µA	0.3222 mW	156.7 MHz	1.578 pA/√Hz	0.8845	111.1 mV
fast/cold	const-g_m	**182.1 µA**	**0.2785 mW**	156.7 MHz	1.578 pA/√Hz	0.8845	111.1 mV

Observation: because $g_m, g_o,$ and all capacitances are *identical* between the two strategies at every corner in this simplified comparison (constant-g_m restores g exactly; g_o and C's use the same multipliers regardless of current per the problem's stipulation), **f_{3dB}, noise, H_0, and V_{DSAT} come out numerically identical** — only **I_D (and therefore power)** differs between strategies. This matches the lecture-deck's constant-g_m bullet ("small variations in GBW… large variations in I_D and V_{DSAT}" — slide 18): here V_{DSAT} was already pinned to the fixed-I_D value by construction of the comparison (the problem fixes $x_c = ax$ for headroom regardless of strategy), so what we actually observe is the current/power swing alone: **+33% current at slow/hot, −17% at fast/cold** relative to nominal, versus the fixed-I_D strategy's flat current (and correspondingly flat power) across corners.

Choice: fixed-I_D biasing is preferable here. It gives identical AC/noise/headroom performance to constant-g_m in this model (since both restore the same relative device operating condition at the margin that matters) while keeping power flat and bounded — constant-g_m trades a 27%-worse worst-case power (0.41 vs 0.32 mW at slow/hot) for no corresponding benefit in this simplified sensitivity experiment. Neither strategy, by itself, fixes the underlying slow/hot H_0/noise/bandwidth shortfall — that required the x, I_D re-optimization already done in Part D.

(Caveat, as stipulated: this is a simplified sensitivity experiment — holding b, d multipliers fixed while only I_D changes — not a full PDK re-lookup at the new bias current; a real constant-g_m bias would move the device to a different point on the real I–V characteristic, changing C_{gg} etc. too.)

SPICE verification plan

1. **Operating point:** confirm M1 V_{GS}, V_{DS}, I_D match the LUT-derived targets; confirm the current-source load delivers exactly I_D at $v_o = 0.60$V with ≥0.15V compliance.
2. **AC transfer:** `.ac` sweep 1 Hz–1 GHz, extract $H(j\omega)$ from v_o/i_S (inject an ac current source at the gate), compare H_0, f_{3dB}, and the RHP zero location against A.2–A.4.
3. **Input-referred noise:** `.noise` analysis referred to the current source at the gate; compare the 100 MHz point against A.5.
4. **Charge-step response:** `.tran` with a narrow ($\ll \tau$) current pulse of area 20 fC into the gate node; compare $v_o(0^+)$, $v_o(\infty)$, and 1%-settling time against A.7; separately confirm the pulse amplitude is small enough that output dv/dt stays below I_D/C_L (no slewing).
5. **Bias headroom:** sweep v_o over 0.60 ± 0.25V, confirm M1 stays in saturation ($V_{DS} \geq V_{DSAT}$) and the current-source load keeps its 0.15V compliance at both swing limits.
6. **Corners:** repeat 1–5 at slow/hot and fast/cold (T, process corner, and supply variation as applicable to the real PDK — not just the stipulated a, b, d, γ multipliers).

Small-signal parameters to extract (replacing the illustrative LUT): `gm`, `gds` (→ $A = g_m/g_{ds}$), `cgg`, `cgs`, `cgd`, `cdb`, and `id/w` at the swept (V_{GS}, L) grid — i.e. regenerate exactly the four LUT columns (x, f_T, A, J) from real BSIM/PDK device data via a `.op` / `.dc` characterization sweep, as in Murmann's slides 7–10.

Real-circuit elements to add: the actual active-load device (replacing the ideal current source — its own g_{ds} adds to g_o and its own noise adds to S_{io}, both currently lumped into the stipulated 0.60 multiplier and γ_{eff}), the real bias-servo circuit (currently idealized as ac-open), the reset switch (adds charge-injection/kT/C noise excluded by Section 5), and extracted parasitic R/C on the gate and output interconnect (beyond the 50fF/75fF lumped estimate used in Part D).

Model limitations (full list)

- LUT is 3 L-points × 5 x-points, interpolated/no-extrapolated — a real PDK characterization would use a much denser grid (see slide 8–9 of the Murmann deck: full $(V_{GS}, V_{DS}, V_{SB}, L)$ sweep).
- Corner model (Section 6) is a *stipulated* multiplicative scaling of nominal g, g_o, C's at **fixed W, L** — not a re-characterization of a real device at the corner's actual bias; $V_{DSAT,c} = 2/x_c$ is an estimate, not an exact saturation boundary (explicitly flagged in the prompt).
- Noise model excludes sensor, switch, flicker, induced-gate, and servo noise — a real implementation needs all of these, especially flicker noise (1/f corner) and reset-switch kT/C noise given the stated exclusions.
- Ideal current-source load (only contributes $0.6 g_{ds}$ and lumps its noise into γ_{eff}) — a real active load has its own g_{ds}, C_{db}, bias sensitivity, and mismatch.
- Linear small-signal model throughout — cannot capture slewing (Part D), large-signal distortion, or switch charge injection.
- Single fixed calibration constant $K_{cal} = 1/H_{0,nom}$ — no per-corner trim, as stipulated; a real system might need temperature compensation if the 5% corner-error budget is tight (it is, at 4.2–4.6% in this design).

Deliverables checklist

- [x] Circuit/testbench diagram, current directions, sign conventions — `gm_id_design.py` header + this report's Section A.1.
- [x] Full derivations with dimensional checks — Part A above.
- [x] Feasible-region plots, all 3 L's — `feasible_regions.png`.
- [x] Current-vs-x tradeoff plot — `current_vs_x_tradeoff.png`.
- [x] Final device/capacitor table, worst-corner margins — Part C, Part D.
- [x] Nominal calibration factor, fixed/const-gm comparison, charge response — Part D, E.
- [x] Model limitations + SPICE validation plan — above.
- [x] Executable script reproducing all of the above — `gm_id_design.py` (run: `python3 gm_id_design.py`).